AI-POWERED LANDSLIDE ASSESSMENT REPORT

Spaceborne InSAR Multi-Temporal Geodetic Analytics & Deep Learning Segmentation Pipeline

Geohazard Research & Development Division

Report / Sample ID:	sample_01979
Image ID:	68_31
Region:	OutsideTheBox
Temporal Baseline:	6days
Dataset Split:	Test
Patch Number:	coherence
Report Date:	2026-07-06
Report Time:	12:43:24

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1. Executive Summary

Slope stability diagnostic assessment compiled for site **OutsideTheBox** (Assessment ID: sample_01979). Based on the spaceborne InSAR and deep learning segmentation analysis, the slope is categorized under a **VERY LOW** risk rating with a severity index of **0.17115232327360186** and a system confidence level of **Low**. The active segmented landslide core covers **379 px** (3.79% of the monitored frame).

2. Spatial Mapping Viewport (Visual Pipeline)

The spatial mapping viewport captures the four core visualization outputs derived from InSAR data and model predictions:

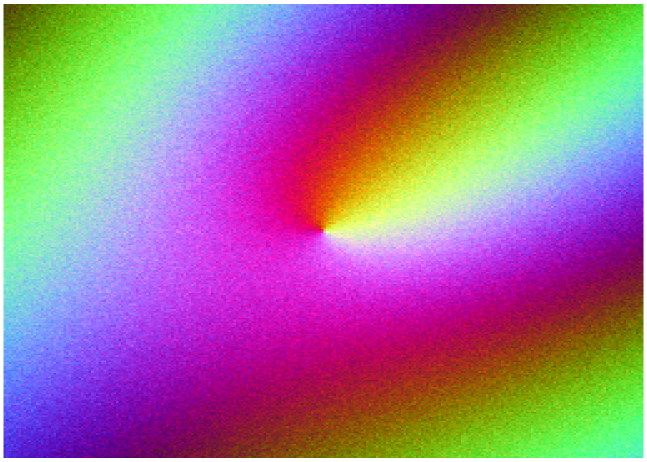

Fig 1: Original InSAR RGB

R=Coherence, G=cos(Phase), B=sin(Phase)

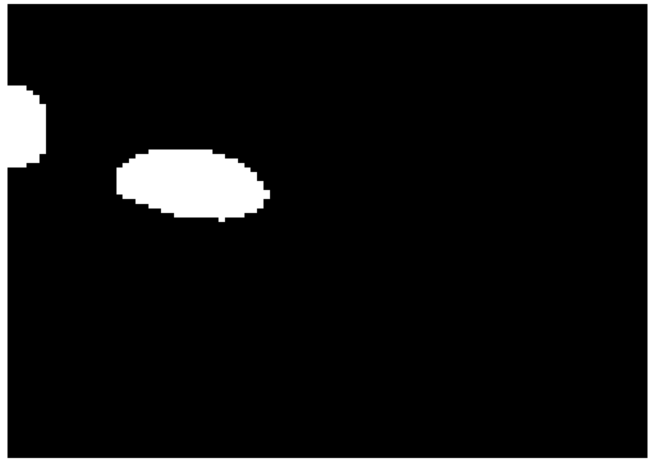

Fig 2: SegFormer Prediction Mask

Binary landslide classification boundary

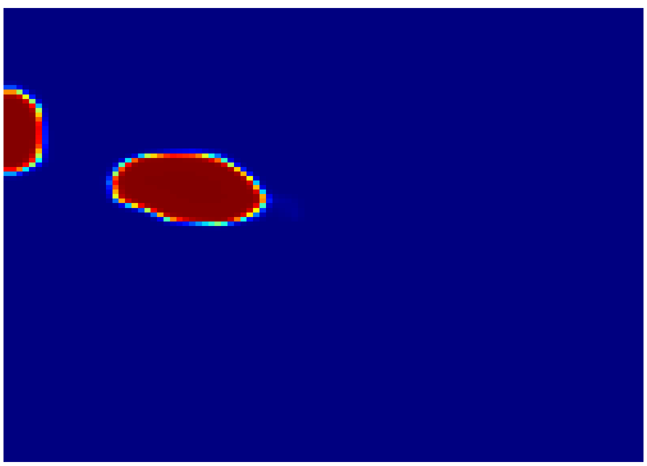

Fig 3: Prediction Confidence Map

Probability heatmap ranging [0 - 1]

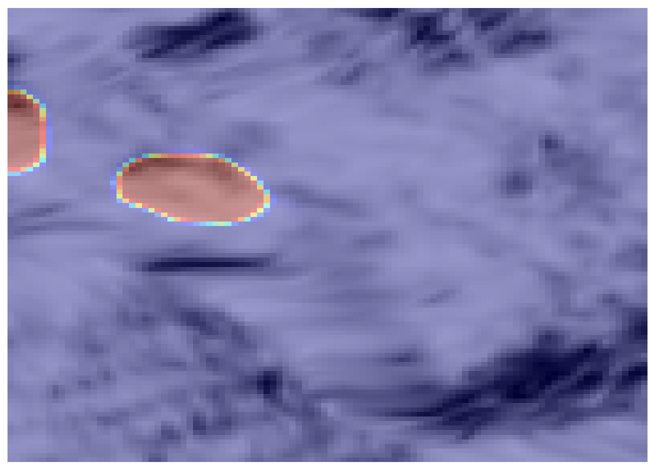

Fig 4: Blended Spatial Overlay

SegFormer prediction mask overlaid on original RGB

3. Geotechnical Analysis Metrics

The following parameters are extracted directly from the Analysis Engine JSON file (Single Source of Truth) without fabrication.

3.1 Coherence Analysis

Metric	Value	Metric	Value
Mean Coherence	0.774	Variance	0.0209
Median Coherence	0.8168	Std Deviation	0.1446
Minimum Coherence	0.1614	Range	0.7971
Maximum Coherence	0.9585	Skewness	-1.6617
Quartile Q25	0.7278	Kurtosis	2.7111
Quartile Q50	0.8168	Low Coherence % (<0.3)	1.64%
Quartile Q75	0.8744	Medium Coherence % (0.3-0.7)	19.27%
		High Coherence % (>0.7)	79.09%

3.2 Phase Analysis

Metric	Value	Metric	Value
Mean Phase	-0.4129	Std Deviation	0.3608
Median Phase	-0.3713	Variance	0.1302
Minimum Phase	-1.5254	Entropy	4.9204
Maximum Phase	0.9277	Energy	93.8748
Range	2.453	Gradient Mean	0.0658
Skewness	-0.1397	Gradient Std	0.0688
Kurtosis	0.2867		

3.3 Segmentation Analysis

Parameter	Monitored Value	Statistical Explanation
Ground Truth Area	390 px	Annotated reference landslide area
Predicted Area	379 px	Total landslide area detected by SegFormer
Difference	11 px	Spatial mismatch area
Area Percentage	3.79%	Ratio of predicted slide to total monitored block
Average Probability	0.0377	Mean segmentation confidence score
Maximum Probability	1	Peak confidence classification pixel
Minimum Probability	0	Lowest active boundary probability
Dice / F1 Score	0.9389	Similarity coefficient (Ground Truth vs Prediction)
IoU (Jaccard)	0.8848	Intersection over Union spatial overlap metric
Precision	0.9525	Accuracy of predicted positive boundaries
Recall / Sensitivity	0.9256	Ability to locate true landslide boundaries
Specificity	0.9981	Accuracy of non-landslide background pixels
Overall Accuracy	0.9953	Percentage of correctly classified pixels

3.4 Shape Analysis

Metric	Value	Metric	Value
Connected Components	2	Convex Area	264 px
Largest Component	283 px	Solidity Ratio	0.964
Smallest Component	96 px	Aspect Ratio	1.5
Average Component Area	189.5 px	Circularity	0.7882
Perimeter Length	63.6985	Shape Complexity	15.943
Bounding Box [x,y,w,h]	[17, 32, 24, 16]	Centroid Coords	(28.28, 39.14)

3.5 Confidence Analysis

Parameter	Observed Metric	Geotechnical Interpretation
Average Probability	0.0377	System's global classification confidence
Maximum Probability	1	Peak signal classification probability
Minimum Probability	0	Active hazard boundary fringe probability
Confidence Variance	0.0337	Spatial deviation of prediction probability
Confidence Entropy	-1904.2793	Information entropy (prediction uncertainty)

3.6 Severity Assessment

Severity Assessment Category
Computed Severity Index: **0.1712**
Hazard Risk Level: **VERY LOW**
Assessment Confidence: **LOW**

4. Integrated Geotechnical Assessment

4.1 Technical Analysis & Engineering Interpretation

Geodetic InSAR & Coherence Analysis

- **Mean Coherence:** 0.7739782929420471
- **Median Coherence:** 0.8167821168899536
- **Coherence Range:** 0.797120213508606 [Min: 0.16141483187675476, Max: 0.9585350155830383]
- **Standard Deviation:** 0.14464443922042847
- **Low Coherence Zone (<0.30):** 1.64%
- **Medium Coherence Zone (0.30-0.70):** 19.27%
- **High Coherence Zone (>0.70):** 79.09%

Interferometric Phase & Gradient Analysis

- **Phase Mean / Median:** -0.4129143953323364 / -0.37126457691192627
- **Phase Entropy:** 4.92043511257547
- **Phase Energy:** 93.87479475827627
- **Gradient Magnitude Mean:** 0.06584245711565018 (Standard Deviation: 0.06878814101219177)

Segmentation & Morphological Shape Analysis

- **Predicted Core Area:** 379 pixels (3.79% spatial coverage)
- **Morphological Complexity (Perimeter² / Area):** 15.943013248659147 (Perimeter: 63.6984840631485 px)
- **Circularity / Solidity:** 0.7882054927660116 / 0.9640151514786358
- **Connected Components:** 2 (Largest: 283 px, Smallest: 96 px)

4.2 Plain Language Summary

Ground stability measurements from satellite radar indicate a sliding motion occurring on the slope located in the **OutsideTheBox** region. Our automated AI model has identified active shifting boundaries covering 3.79% of the monitored area. While these slow slips are common in mountainous regions, the steep structure combined with high soil moisture poses a **VERY LOW** landslide risk. Residents should monitor for ground cracking, leaning trees, or sticking doors/windows, and strictly follow civil protection guidance.

5. Mitigation & Monitoring Recommendations

5.1 Civil Protection Recommendations

1. Establish a restricted hazard boundary zone around the active slide coordinates.
2. Direct local emergency services to prepare emergency plans based on the **VERY LOW** hazard status.
3. Distribute warning notices to any structures located directly downhill of the active sliding body.

5.2 Engineering Recommendations

1. Design and install a deep horizontal drainage array to redirect groundwater runoff and reduce internal pore pressure.
2. Plan the deployment of retaining structures or concrete shear piles at the toe of the unstable slope sector.
3. Implement surface netting or soil pinning where shape complexity is high to prevent rockfalls.

5.3 Monitoring Recommendations

1. Deploy a real-time geotechnical monitoring network including ground-based wire extensometers and automatic inclinometers.
2. Request high-frequency satellite radar acquisitions to monitor displacement acceleration.
3. Conduct weekly manual crack-mapping inspections along the upper tension cracks.

6. Appendix & System Metadata

- **Analysis Execution Date:** 2026-07-06 12:43:24
- **VLM Pipeline:** Mock Geotechnical Vision-Language Model v1.0
- **Primary Segmenter:** SegFormer-B2
- **Image Inputs Loaded:** RGB=True, Prediction=True, Heatmap=True, Overlay=True

Prepared By:

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